

Screening peer-lending loans for better outcomes¹

Lending Club (www.lendingclub.com) started as a peer-to-peer lending platform that set up an online marketplace connecting investors to borrowers. Their stated goal was to “transform the banking system to make credit more affordable and investing more rewarding. [They] operate at a lower cost than traditional bank lending programs and pass the savings on to borrowers in the form of lower rates and to investors in the form of solid returns.” For a description on how it worked, check out [this archived website](#). It has since moved on to become a broader banking service. The rest of the case is written from the vantage of its earlier years as a peer-to-peer lending platform.

A key asset of the platform is the information it collects in loan applications and the resulting outcomes such as the rejection of loan applications, or in the case of accepted loans, the terms of the loan (such as the duration and interest rate). Finally, the platform also captures all the information about the actual payments from the borrower and whether any of the loans were fully paid back. The latter could happen if the borrower defaulted or the loan was charged off, with the borrower deemed unlikely to pay it back (which is essentially a virtual default).

Lending Club has approached your firm for advice on how this key data asset can be used to screen loans applications received by the platform while being presented to investors. **The main goal is to reduce the risk to Lending Club of featuring loans in the platform that might eventually default.** For this purpose, your team has been provided with a small data sample of historical 3-year loans that were funded, with a good representation of loans that were not paid back in full (see the Loans Data Description file accompanying the data). The cleaned data file they have supplied is loans-kpi.csv that comes with an accompanying data description file LoansDataDescr.pdf.

Your task is to devise an end-to-end pipeline of analysis to arrive at recommendations for the Executive Committee of Lending Club on using this data to refine the loans they fund. **The CEO is looking to increase the return on invested capital for the platform users**, the **CFO is focused on profits earned in the form of interests over the loans** (of which you may assume that Lending Club partakes a cut), while the **CMO is focused on growing the market share of Lending Club against that of competitors** like Prosper. These three CXO's form the audience for your analysis, and your goal is to **devise a method to screen the loan applications made** to the platform for their risk to each of these concerns above. You should use the suggested questions below to generate code in colab to help you with the three phases of the process; if you prefer, you can get the code to output Excel files for further analysis. The last phase involves consolidating all your analysis for effectively communicating your recommendations and insights and transferring over a prototype folio of the models and analyses on which you have based your recommendations.

¹© R. Ravi 2026. This case was developed by Professor R. Ravi based on data that used to be publicly available from Lending Club. The situation described, though based on real data, is developed only for the sake of illustration. The data set used here is adapted from the online resources for the book The Analytics Edge by Bertsimas, O’Hair and Pulleyblank (Dynamic Ideas LLC, 2016).

Update 1: Diagnostic and Descriptive Models

The goal of this first update is to describe your business understanding and data understanding of the problem. Articulate what business problem(s) you are trying to formulate and solve with the data. Based on your descriptive analysis, demonstrate your understanding of the provided data and how this understanding might help you upstream in understanding your loan applicants' characteristics and modeling their default behavior. Deepen your understanding of how the different variables are correlated and vary with each other. **Clustering allows you to use these co-varying patterns to pick out similar sets of loan applications. PCA highlights weighted combinations of the variables representing directions of highest variability among the data points** given to you.

For this and later updates, use the questions below to generate code in colab via generative AI prompts to carry out your analysis. Structure your report as an update provided to the ExCo client at this early part of the project. Use the following questions and your work to structure your presentation.

- Business Understanding: What decisions are the ExCo trying to make (e.g., which loan applications to feature to investors, how to price interest rates by risk tier, or which borrower segments to prioritize for growth)? Which of these decisions of Lending Club will you try to supply recommendations for?
- Source of value: How could you use past data to help make decisions for Lending Club (e.g., using the historical record of funded loans and their outcomes to flag high-risk applications before they are ever shown to investors)?
- Data Understanding: Build a high-level description of the different 'features' — the variables describing the loans. Which do you think are most important for Lending Club in determining its eventual profitability for the platform (e.g., credit score, interest rate, debt-to-income ratio, revolving utilization)? Why?
- What is the unit of analysis in the data (i.e. what does one row represent)? What is the data dynamism (which fields are known at the time of application versus computed only after the loan's outcome is known, such as tot.payment, principal, and interest)? Are there highly correlated variables (e.g., interest rate and FICO score, or revolving balance and revolving utilization)?
- Run basic descriptive statistics: did you find any interesting relationships that are relevant to finding loans that might default (e.g., how default rates vary with credit score, interest rate, loan purpose, or debt-to-income ratio)?
- Run a K-means analysis on the scaled numerical variables in the data (e.g., interest rate, installment, annual income, FICO score, revolving balance, revolving utilization) and describe the results.
 - Use your understanding of the variables to determine a set of variables for carrying out the clustering.
 - Determine an appropriate number of clusters to explore more carefully.
 - For the decided number of clusters, use visualizations of the cluster centers (e.g. by using parallel-lines plots) to "name" each cluster based on its distinguishing features (e.g., "high-FICO low-rate safe borrowers," "high-utilization revolvers," "short-history small-purchase borrowers").
 - How could you use this clustering of the loans to guide future work on modeling?
- Perform and interpret the outputs of a principal component analysis of the numerical variables in the data.
 - Use your understanding of the variables to determine a set of variables for carrying out the PCA.
 - Interpret the principal components using a model-driven analysis by examining the various variable loadings.
 - Interpret the principal components using a data-driven analysis by examining some extreme data points in the biplot of the top principal components.
 - How could you use this PCA of the loans to guide future work on modeling?

Update 2: Predicting Default Using Logistic Regression and Trees

The goal of this second update is to run and interpret classification models based on logistic regression and decision trees. As before, use the hints provided below to structure your code-generating prompts and your analysis. Structure your report as an intermediate update provided to the ExCo client. Use the following suggested questions.

- Logistic regression model:
 - Examine the correlations between the available variables (e.g., interest rate vs. FICO score, or revolving balance vs. revolving utilization) and choose a subset of the variables to finalize and include in your model. Describe how you arrived at this set.
 - Run a logistic regression with the selected variables; summarize your findings.
 - Visualize your model on a few variables (e.g., FICO score, interest rate, debt-to-income ratio) to better understand the relationship of the output to the different important variables.
 - Interpret the coefficients and their signs and build a verbal explanation of the model (model-driven explanation). Order the significant variables in decreasing rank of importance.
 - Pick a few sample loans that score very high or low and use their variable values to explain the model (data-driven).
- Decision tree model:
 - Run a decision tree model on a training data set. Interpret your results.
 - Down sample the default loans to make a new data set that has an equal number of defaulters and non-defaulters. Build a decision tree on this down-sampled data. Which of these models will you retain and why?
 - Should you worry about which variables to retain in building a tree model by examining the correlations between them (e.g., revolving balance and revolving utilization)? Why or why not?
 - Interpret some of the key rules in the decision tree you finalize (e.g., a low FICO score combined with a high debt-to-income ratio). Do they make logical sense to a lending analyst?
- Preliminary model comparison:
 - Compare the performance of your best tree model and your best logistic regression model.
 - Consider how you might use the two models to screen loans. Do the qualities of the predictions change as you change the cutoff threshold to signal default in deploying your model predictions?
- (Optional) Build a random forest model to see how it compares with the other two predictive models.

Update 3: Prescriptive Models using Optimization and Communicating Your Results

The goal of this week's update is to make a final choice of which type of model you will use for the predictions (logistic regression or a tree/random forest model). Once you decide on this, you should next pick the threshold (score cutoff) at which to deploy this model. Once you have decided which model to use, you can examine the full list of predicted default probabilities from that model and use a more refined metric from these predictions to choose the threshold from the spreadsheet of these predictions for your full test set. Since this is your last update, please spend some effort to frame your final recommendation convincingly, intuitively explain the salient features of the model you recommend for loan screening, and call out any risks in using your model.

- Model and variable choice:
 - Not all variables in the data set may be available at the time of loan application (e.g., tot.payment, principal, and interest are computed only after a loan's outcome is known and should be excluded from the funding-decision model). Construct your final model on an appropriate set of variables to see how well your model does.
 - In choosing between the models, think carefully about which model is both accurate and explainable to the ExCo.
- Threshold choice:
 - To evaluate the effectiveness of different thresholds, consider the concerns of the different members of the ExCo and how these cutoffs help mitigate the risk concerning each of them. If possible, come up with an optimization framework for choosing the optimal threshold for your chosen model.
 - Use the output of the default probability predictions from the chosen model to cast the threshold decision problem as maximizing a desirable output (like return on invested principal, net profit, or market share) subject to constraints on the other outputs (e.g., a minimum acceptable approval rate or a cap on the aggregate risk of the featured loan portfolio). (Output these predictions to an excel spreadsheet if you find it more convenient.) Set up this optimization problem and use this to come up with the final screening recommendation using the chosen model.
 - Alternately, rather than decide on a specific threshold for each model, examine how various metrics that different members of the ExCo may care about (return on invested principal, net profit, approval/market-share rate) change with differing screening thresholds. Try to visualize the metrics in a way that will aid the ExCo to arrive collectively at an appropriate threshold.
- Final recommendation presentation and Risk identification:
 - Structure your recommendation in a top-down manner with the potential gains from your analysis first and then substantiate your model and claims.
 - Explain both the choice of the screening threshold and the model you propose to screen the loan applications in ExCo-friendly terms. For the former, you may try to lead through an example. For the latter, consider using intuitive visualizations or explanations of your chosen predictive model (such as regression coefficients or decision rules).
 - Give a simple explanation of the working of your chosen prediction model to build confidence in your audience of the robustness of your model recommendations.
 - List the key assumptions that must hold for your claimed benefits to hold (e.g., a stable interest-rate environment, no material shift in the risk profile of applicants). Suggest further analysis to mitigate any risks in deploying the model in the platform. Call out any data limitations that might have potentially skewed the results of the models on which you are basing your recommendation (e.g., the sample covers loans funded between 2007 and 2010 and may not reflect current underwriting standards or economic conditions).